

Crash Course in Fast Food: Executive Summary

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1 Introduction

As frequent pedestrians and cyclists, the city of Columbus, OH, does not always feel safe to us. This is especially true during rush hours on weekdays, where many cars are on the road. Living in a busy urban area where lots of fast food chains are located conveniently for people driving in and out of town, anecdotal evidence seems to indicate that our most dangerous car encounters happen when people drive in and out the drive-through areas of the fast food restaurants. We believe that the large speed difference between cars on the road and cars entering the road leads to many dangerous situations. Likely, drive-through costumers are on a slightly higher stress level than drivers who are not trying to optimize their time for food intake.

Understanding if and how the surrounding environment (traffic lights nearby, blind corners etc.) influences the possibility of accidents close to the drive-through restaurants can help improving road safety for all participants. As the city of Columbus is growing, the information can be useful when developing new neighborhoods with a mix of residential and commercial areas. Evaluating if certain commercial vendors need extra signage, more space, or maybe even a redesign of the entrance and exit driveways will help plan the development of future mixed residential/commercial areas.

We will not focus on general accident causes or general road improvements. Our analysis should only include locations of fast food restaurants, as these are places where we understand incoming and outgoing traffic and these places can be easily located on online maps, as all restaurant chains have marked their locations online.

2 Dataset and Stakeholders

We requested the crash data from the Ohio Department of Public Safety, and received a CSV file containing information on 1.289.913 incidents within the time frame from September 2019 to September 2025, covering the whole state of Ohio.

Our analysis can help developers, road planning offices, and the franchise owners alike, making the restaurant experience not only pleasant, but also safe. The Ohio Department of Public Safety is another stakeholder, as the results can inform their road safety agenda.

3 Preprocessing

The dataset includes accidents involving cars in the whole state of Ohio. We thus extract a dataset only for Franklin county. We furthermore focus on weekdays and exclude public holidays, as routines often differ which could blur any patterns we detect.

4 Analysis Methods

We want to compare the drive-through aspect of “getting food” with places that require more time, i.e. a pizza restaurant. Although these places are often located at similar spots than the fast-food chains, pizza restaurants usually do not have a drive-through option, as preparing pizza takes too long to be efficient for customers waiting in their cars.

We perform a Geographically Weighted Regression (GWR) that evaluates if an event at a certain location is more likely than its statistical distribution. The GWR model takes into account that local coefficients can vary spatially, identifying regional differences in the strength and direction of the observables. Applying this model, we want to identify potential hotspots with above-average correlation between crash sites and fastfood chain locations using the following equation:

$$D_{\text{Crash}}(x, y) = a(x, y) + b(x, y) \cdot D_{\text{Fastfood}}(x, y) + c$$

with the densities D for crash events and fastfood restaurant locations, the spatial coordinates (x, y) , represented by longitude and latitude values, and $a(x, y)$ the intercept term of the regression giving us the expected value for no correlation at a specific location (x, y) . With $a(x, y)$ we can describe naturally occurring higher densities in the crash map (e.g. a narrow curve, a blind turn), that have nothing to do with the location of fastfood restaurants. Parameter $b(x, y)$ hints at the strength of the relationship between car crash location and fastfood restaurant location.

With the residual c we can identify weaknesses of the model or other irregularities.

5 Results

6 Future Goals

The comparison between different types of food venues can be taken further to other types of shops near the road. We could use coffee shop locations and identify if dinner and caffeine cravings lead to different levels of attention. To take out the food component completely, we can compare the crash locations to gas stations nearby. Their design is similar to restaurant drive-thrus, but since the main focus of a gas station is fuel and not dinner, the mood of the costumers might be very different.